

Introduction to HCI

Lecture 1

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Humans

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A HUMAN

Has various ways of perceiving the world around him



Seeing



Feeling



Hearing



Smelling & Tasteing





A HUMAN

Also has





Humans

Humans possess more than just senses; they also have memories, experiences, skills, and knowledge.

In HCI, it's crucial to **consider all aspects of the human experience**, including how individuals perceive and interact with their environment, as well as their history of using computers and technology.

It is important to understand **these elements in designing effective and user-friendly interfaces**.





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Computers

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COMPUTER

Cars

Doors

Game Controllers

Toys

Oven

AR Pokemon





Computers

What constitutes a computer?

We typically think of various devices as computers, such as desktops and laptops. Then it expands the definition to include mobile devices, augmented reality systems, and even everyday objects that incorporate computing technology.

Human-Computer Interaction (HCI) is pervasive in our daily lives, especially with the rise of mobile devices and applications like augmented reality games, such as Pokémon Go.

HCI is not limited to traditional computers but encompasses all interactions we have with technology in various forms.





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Interaction

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Interaction

The nature of interaction between humans and computers within the context of Human-Computer Interaction (HCI) can be viewed in two ways:

1. The direct interaction between the human and the computer, where they communicate back and forth.
2. A more nuanced perspective where the human interacts with a task through the computer, with the computer acting as a mediator.



INTERACTION



HUMAN



COMPUTERS



TASKS



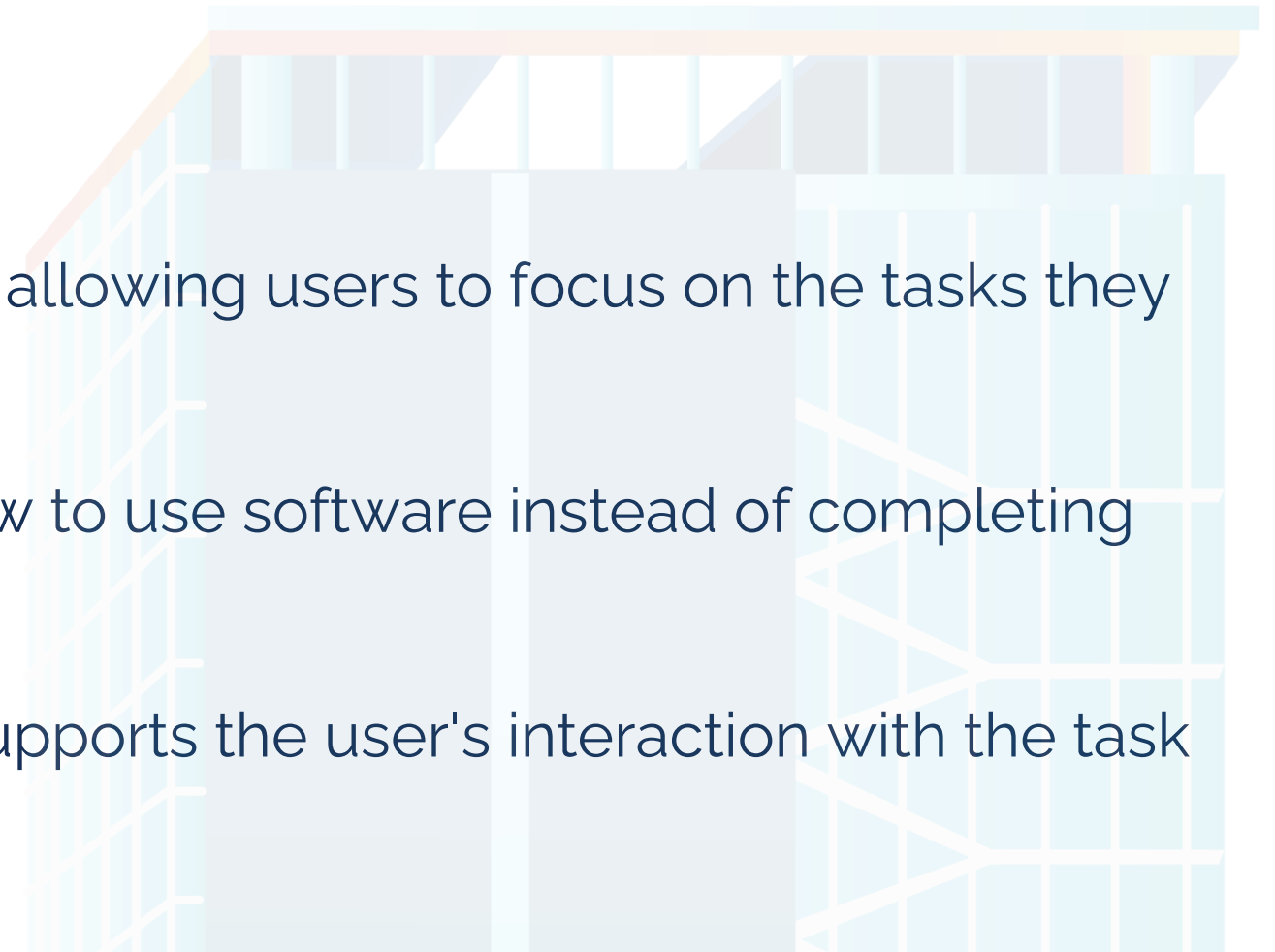


Interaction

The goal of HCI design is to make the interface as invisible as possible, allowing users to focus on the tasks they want to accomplish rather than the interface itself.

Users often feel frustrated when they spend more time figuring out how to use software instead of completing their intended tasks.

Designers aim to create experiences where the interface seamlessly supports the user's interaction with the task at hand.



HUMAN



COMPUTERS



TASKS

Exercise

Give an example where you spend most of your time thinking about the task and an example where you spend most of your time thinking about the tool.

The HCI Space

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HCI scope and significance

While many people believe they are experts in interacting with computers due to their daily use, true expertise in designing interactions for others requires a deeper understanding of HCI principles.

Individuals often underestimate the complexity of HCI. Initially, they may feel confident in their knowledge, but as they learn more, they realize the vastness of the field and the depth of what there is to know.



Part of HCI

UBIQUITY

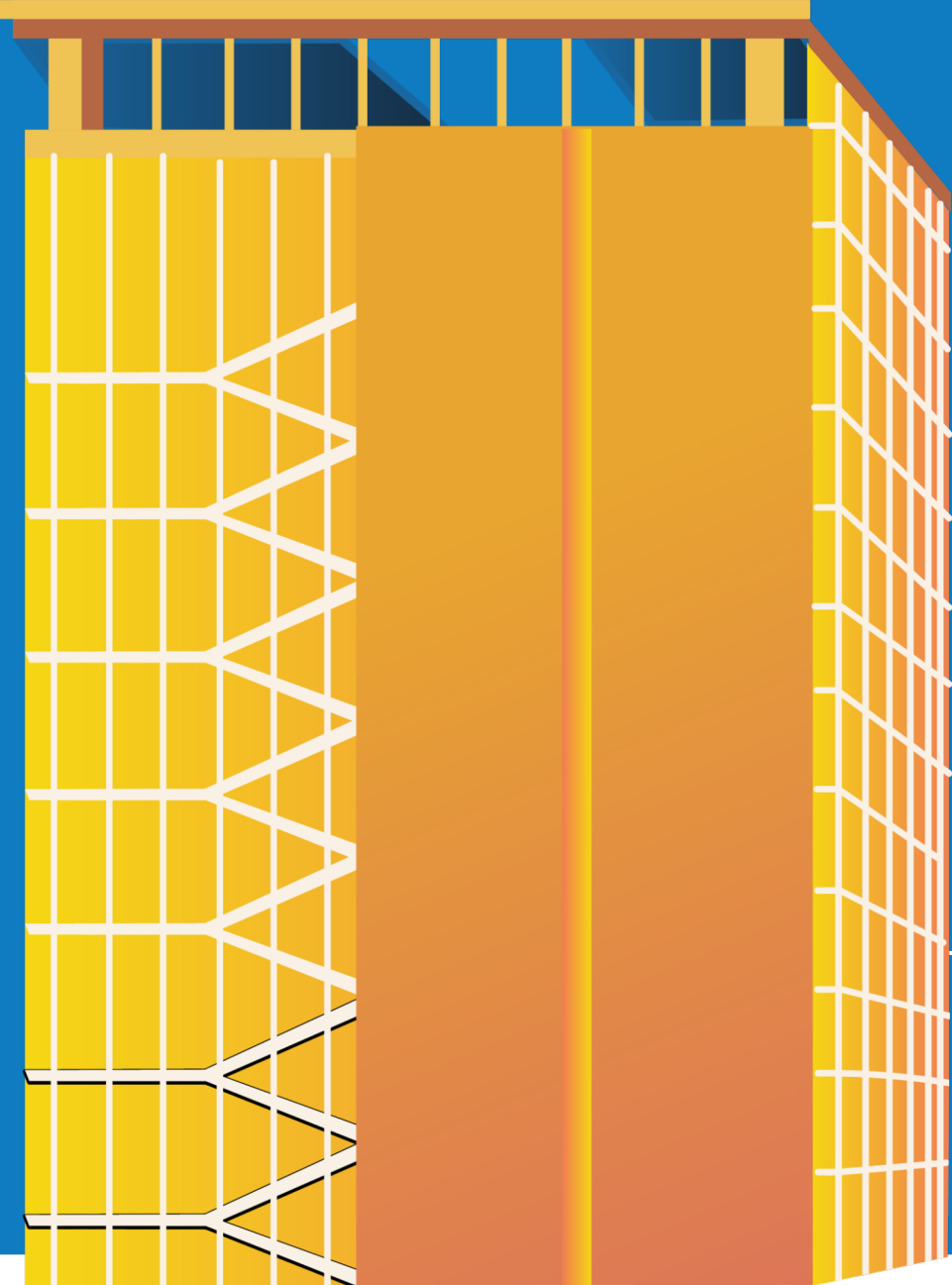
The state or capacity of being everywhere

Challenges of HCI



Human are expert in using computers daily

“We might be experts at interacting with computers, but that doesn’t make us experts at designing interactions between other humans and computers.”

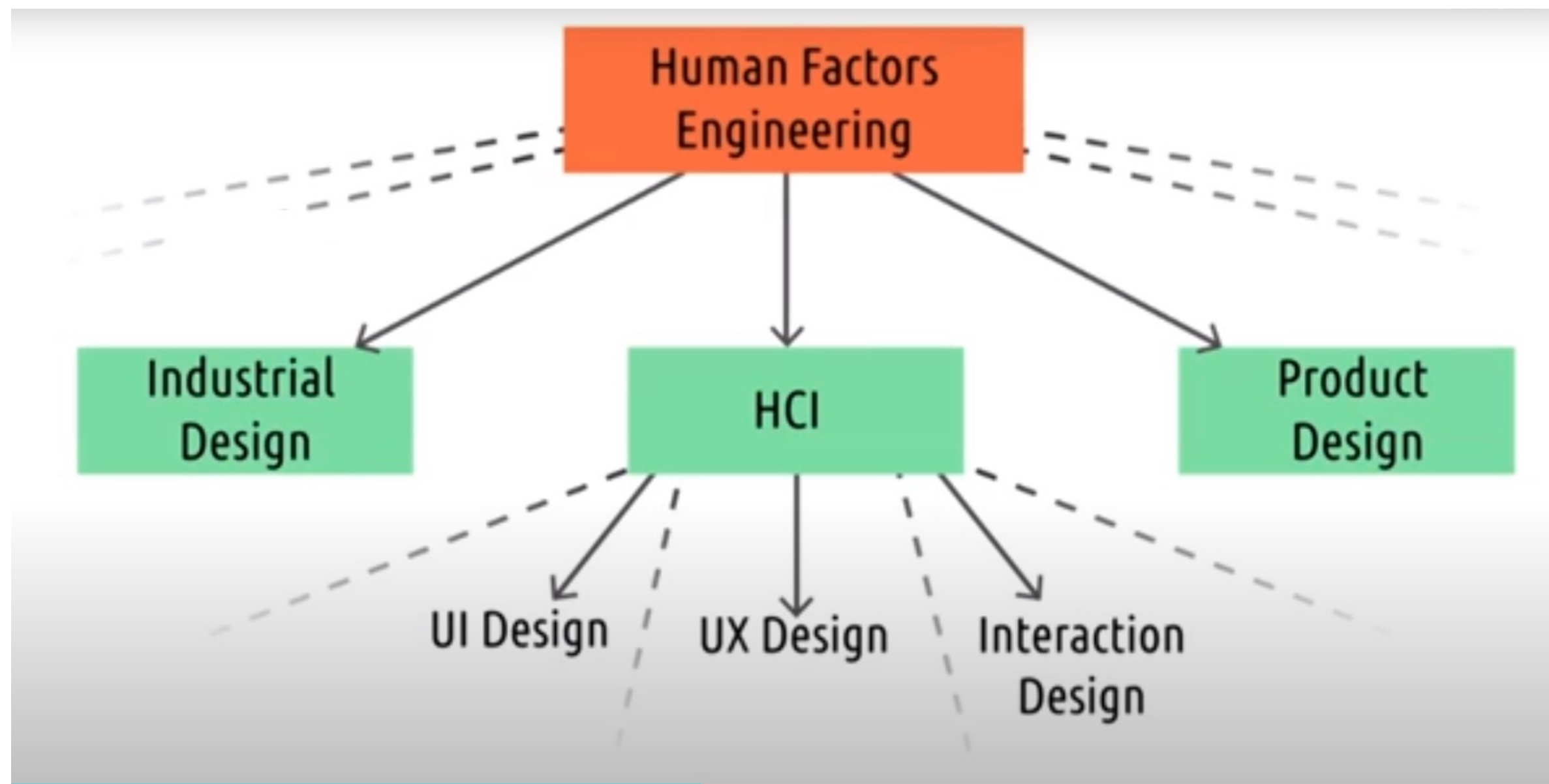


HCI in the Big Picture

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HCI in Big Picture





HCI in the Big Picture

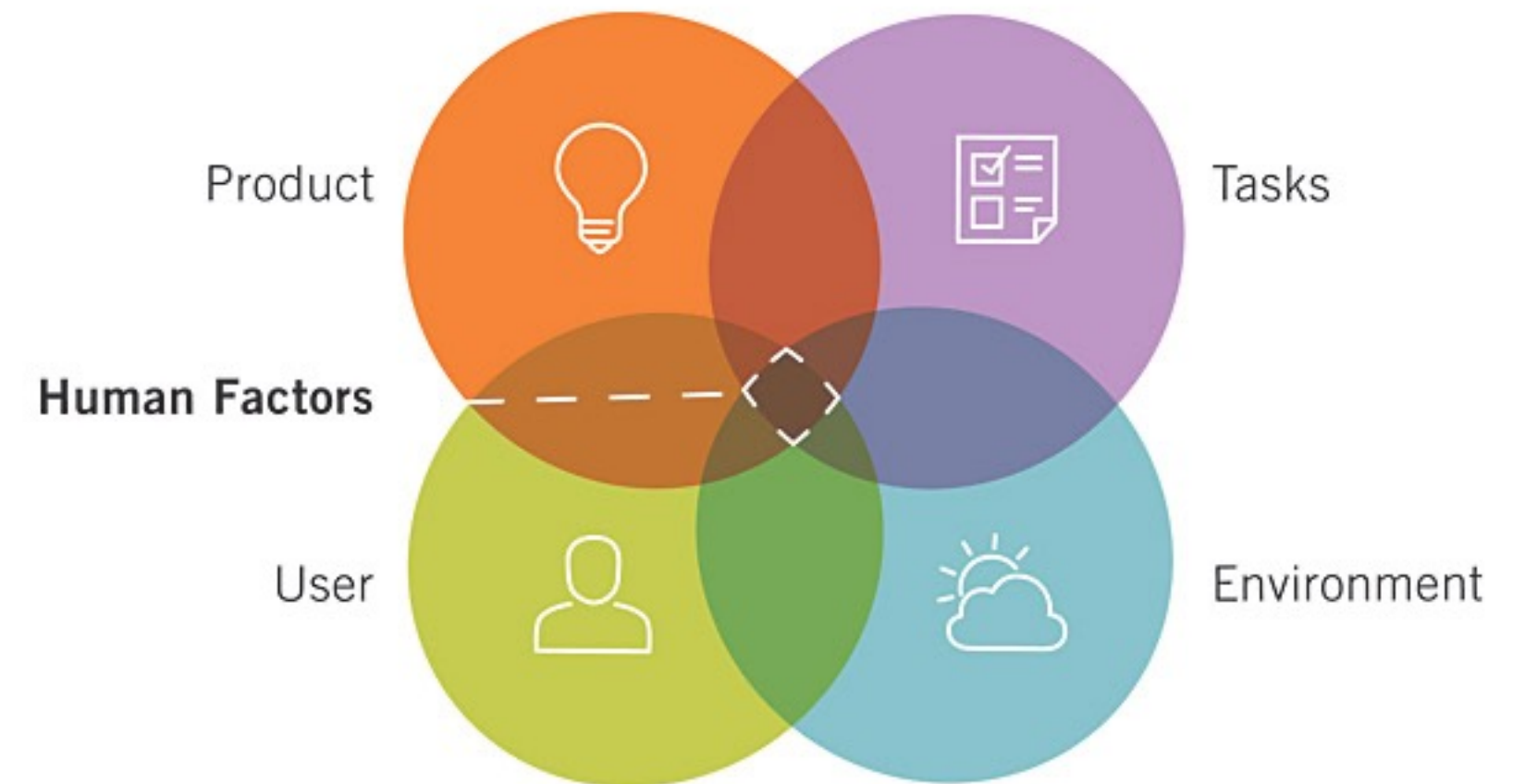
- HCI is a subset of the broader field of human factors engineering, which encompasses various disciplines focused on understanding how humans interact with systems, not just computers
- There are different sub-disciplines within HCI, such as user interface (UI) design and user experience (UX) design, and how they relate to one another.
 - While these areas share principles with HCI, they may apply them in narrower contexts or with additional methods
- HCI connects to other fields, and it is important to design effective and user-friendly systems



HCI vs Human Factor

Human Factor Engineering

Design interactions between people and products, systems, or devices





HCI vs. Human Factors

- Human factors focus on designing interactions between people and various products, systems, or devices, which includes but is not limited to computers
- Whereas HCI specifically examines interactions between humans and computers, human factors also consider non-computerized elements, such as the physical design of product
- As technology becomes more integrated into everyday objects, the lines between HCI and human factors are blurring.
- There is a growing relevance of HCI in diverse applications as computers become ubiquitous in our lives.





HCI vs UI Design

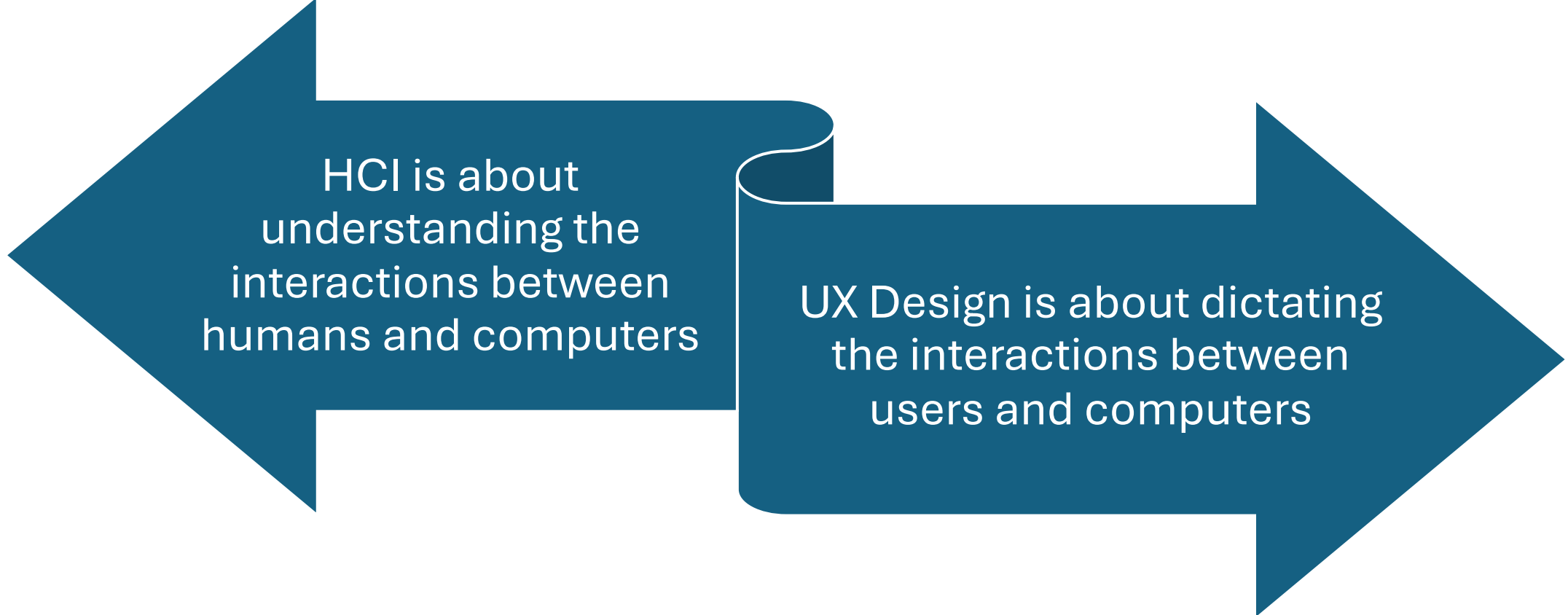
HCI

Mote on things that go beyond the user's interaction with a single screen

UI Design

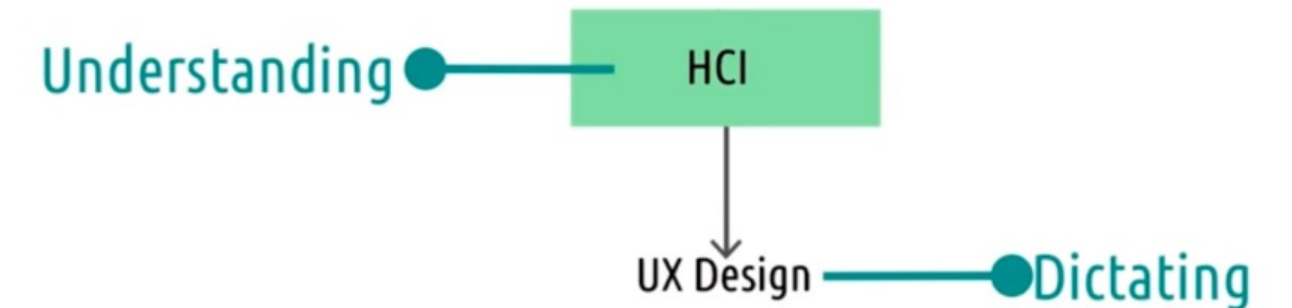
Focus on screen interaction

HCI vs UX Design



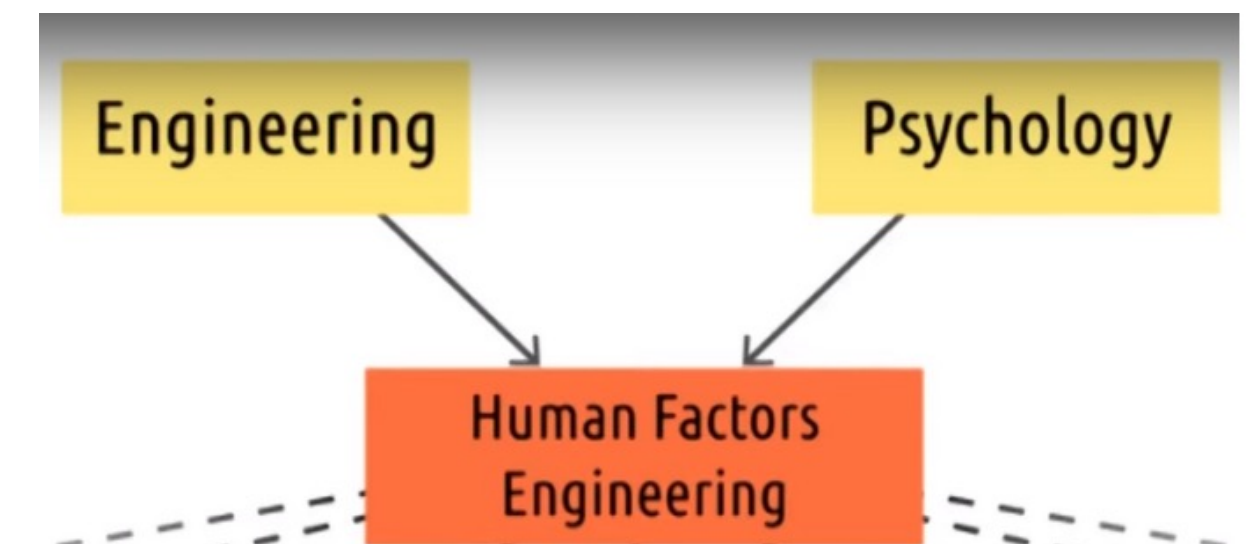
HCI is about
understanding the
interactions between
humans and computers

UX Design is about dictating
the interactions between
users and computers

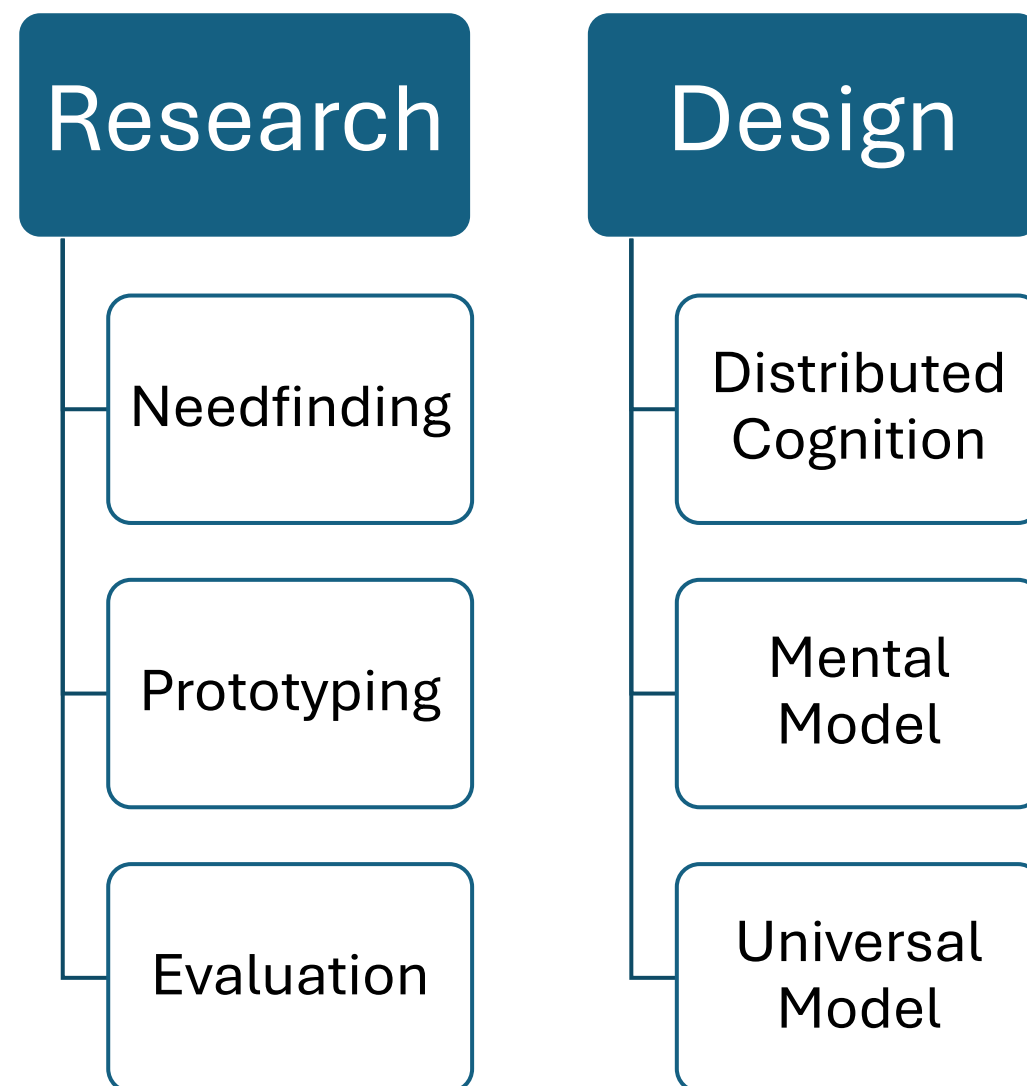


HCI vs Psychology

- We use our understanding of psychology, of human perception, of cognition to inform the way we design interface
- We then use our evaluations of those interface to reflect on our understanding of psychology itself



HCI Research and Design



HCI New Application Areas



TECHNOLOGIES

emerging technological capabilities that let us create new and interesting user interactions.



DOMAINS

pre-existing areas that could be significantly disrupted by computer interfaces like healthcare and education.



IDEAS

the theories about the way people interact with interfaces and the world around them.

Technology: Virtual Reality



an immersive computer-simulated reality

Technology: Augmented Reality



Real-world environments that are complemented by computer-generated multimedia

Technology: Mobile



Portable computing devices built to be used easily on the go

Idea: Context Sensitive Computing



Equipping user interface with historical, geographical, or other forms of contextual knowledge



Domain: Special Needs

Domain: Education

Domain: Healthcare

Assignment

Choose an area: Technology, Idea, or Domain, then claim an example case in the area to be explored and presented briefly





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Recap: Welcome to HCI!

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Summary

- provides an overview of what students can expect to learn in the Human-Computer Interaction (HCI) course
- outlines key topics, such as fundamental design principles, the importance of user research for developing new interfaces, and the interplay between research and design
- highlights how research informs design decisions and how design can facilitate further research
- mentions the application of HCI principles across various domains, including augmented reality and healthcare





Thank You

Tim RMK GIGA

